

ON PROBABILITY DISTRIBUTIONS FROM MIXTURES OF MULTIVARIATE DENSITIES

W.Y. Tan

Department of Math. Sciences, Memphis State University, Memphis, Tennessee 38152, U.S.A.
and Oak Ridge National Laboratory, Oak Ridge, Tenn.*

Key words: Invariance, mixtures of multivariate densities, permutation, sample covariance matrices.

SUMMARY

This paper provides some sampling distribution results of statistics from mixtures of continuous multivariate densities. Specifically the present paper derives the joint sampling distribution of statistics which are invariant with respect to permutations on observations from mixtures of continuous multivariate densities. In Section 4, results of sampling distributions of linear and quadratic forms from mixtures of multivariate normals are also derived. These results are then applied to derive joint null and nonnull distributions of sums of squares matrices from multivariate regression models.

1. INTRODUCTION

In multivariate statistical analysis a vital element is the assumption of normality for the random variables. In practice, however, it is often the case that the random variables are in fact not normally distributed: (see for example, Ito (1968), Layard (1972, 1973, 1974), Mardia (1971, 1975), Davis (1976), Lachenbruch et al. (1977), Krzanowski (1977), Waternaux (1976, 1977) and Tan (1978), among others). When such is the case, how would the departure from normality affect the conventional statistical inference procedures? Specifically, one may wonder what would be the sampling distributions of some commonly used statistics? For providing some answers to the above question, Tan (1978) derived the distribution of sample covariance matrix by assuming that the random vectors follow a mixture of multivariate normal distributions with a common covariance matrix. The present paper provides some general results for statistics from mixtures of multivariate densities and mixtures of multivariate normal densities. This would provide extensions of results given in Tan (1978). We note in passing that, while it is a mathematical convenience to choose a mixture of densities and mixtures of normals for the random variables, the mixtures of normals are in fact very common in medical, biological and engineering experiments, see for example, Hyrenius (1950, 1952), Subrahmaniam (1972), Subrahmaniam et al. (1975), Box and Tiao (1968), Tiao and Ali (1971), Tan and Chang (1972), Tan (1975), Lee and D'Agostina (1976) and Lee and Gurland (1977), etc.

*Operated by Union Carbide Corporation under contract W-7405-eng-26 with the U.S. Department of Energy.

2. Some Preliminaries and Notations

We denote by $\Pi_k(p_i; q_i(\mathbf{x}))$ the universe which is a mixture of k populations with densities $q_i(\mathbf{x})$ and mixing proportions p_i , $i=1, \dots, k$, i.e., the density function for the universe is

$$\sum_{i=1}^k p_i q_i(\mathbf{x}).$$

As in Tan (1978), we let δ_{ij} , δ_{ij} and $\delta_{ij\ell}$ be nonnegative integers taking values 0 and 1 only, and let a_i , s_i , s_{ij} and n_j be nonnegative integers. Furthermore, the following notations will be used throughout the paper unless otherwise stated:

$\sum_S$ denotes summation over all elements in the set S ;

$$\binom{n}{s_1, \dots, s_k} = (n!) / \{s_1! \dots s_k!\}, \text{ where } \sum_{i=1}^k s_i = n, k > 1;$$

$$h(s_{ij}, s_i; n_j) = h(s_{ij}, s_i; n_j, i = 1, \dots, k, j = 1, \dots, m)$$

$$= \left\{ \prod_{j=1}^m \binom{n_j}{s_{1j}, \dots, s_{kj}} \right\} / \binom{N}{s_1, \dots, s_k},$$

where $\sum_{j=1}^m n_j = N$, $\sum_{i=1}^k s_{ij} = n_j$, $j = 1, \dots, m$ and $\sum_{j=1}^m s_{ij} = s_i$, $i = 1, \dots, k$

(we write simply $h(s_{ij}, s_i; n) = h(s_{ij}, s_i)$ if $n_j = n$ for all $j = 1, \dots, m$);

$W_k(s_i; N)$ denotes the set of all k -tuples $(s_1, \dots, s_k)$ satisfying $\sum_{i=1}^k s_i = N$;

$H_{k,N}(\delta_{ij}, a_i)$ denotes the set of all kN -tuples $(\delta_{ij}, i = 1, \dots, k, j = 1, \dots, N)$ with elements δ_{ij} satisfying

$$\sum_{i=1}^k \delta_{ij} = 1, j = 1, \dots, N, \quad \sum_{j=1}^N \delta_{ij} = a_i, i = 1, \dots, k$$

and

$H_{k,m,n}(\delta_{ij\ell}, s_{ij}, s_i)$ the set of all kmn -tuples $(\delta_{ij\ell}, i = 1, \dots, k, j = 1, \dots, m, \ell = 1, \dots, n)$ with elements $\delta_{ij\ell}$ satisfying

$$\sum_{i=1}^k \delta_{ij\ell} = 1, j = 1, \dots, m, \ell = 1, \dots, n, \quad \sum_{\ell=1}^n \delta_{ij\ell} = s_{ij}, i = 1, \dots, k, j = 1, \dots, m \text{ and}$$

$$\sum_{j=1}^m s_{ij} = s_i, i = 1, \dots, k \text{ (Note that } \sum_{\ell=1}^n \left\{ \sum_{i=1}^k \delta_{ij\ell} \right\} = n \text{ so that}$$

$$\sum_{i=1}^k s_{ij} = n, j = 1, \dots, m \text{ and } \sum_{i=1}^k s_i = mn).$$

In this paper we shall also let $x \sim N_r(\mu, \Sigma)$ denote that the $r \times 1$ random vector x is distributed as multivariate normal with mean μ and covariance matrix Σ and $S \sim W_r(f; \Sigma, \theta)$ that the $r \times r$ symmetric random matrix S is distributed as noncentral Wishart with d.f. = f , parameter matrix Σ and noncentrality θ (we write $S \sim W_r(f; \Sigma)$ for central Wishart); the densities of $x \sim N_r(\mu, \Sigma)$ and $S \sim W_r(f; \Sigma, \theta)$ will be denoted, respectively, by $N(x; \mu, \Sigma)$ and $G_r(S; \Sigma, \theta)$.

In the above notations, one notes immediately that $h(s_{ij}; s_i; n_j) = 0$ if $s_{ij} > n_j$ for some (i, j) ; further, if $g(s_{ij})$ is a function of s_{ij} , $i = 1, \dots, k$, $j = 1, \dots, m$, then

$$(2.1) \quad H_{k,m,n}(\delta_{ij}; s_{ij}; s_i) \quad g(s_{ij}) = W_k(\Sigma; s_i; mn) \left\{ W_k(\Sigma; s_{i1}; n) \cdots W_k(\Sigma; s_{im}; n) \right\} g(s_{ij}) \\ \times \left\{ \prod_{j=1}^m \binom{n}{s_{1j}, \dots, s_{kj}} \right\} = W_k(\Sigma; s_i; mn) \left\{ \binom{mn}{s_1, \dots, s_k} \right\} \left\{ W_k(\Sigma; s_{i1}; n) \cdots W_k(\Sigma; s_{im}; n) \right\} \\ g(s_{ij}) h(s_{ij}; s_i),$$

The following two lemmas, given in Tan (1978), will be used in deriving some of the results.

Lemma 1

$$(2.2) \quad \prod_{j=1}^n \left\{ \sum_{i=1}^m b_{ij} \right\} = W_m(\Sigma; n) H_{m,n}(\delta_{ij}; s_i) \left\{ \prod_{i=1}^m \prod_{j=1}^n b_{ij} \delta_{ij} \right\},$$

Lemma 2

Let X be a $n \times r$ matrix of random variables whose rows x_i' are independently distributed as $x_i \sim N_r(d_i, V)$, $i = 1, \dots, n$.

Let A_j , $j = 1, \dots, p$, be symmetric $n \times n$ matrices satisfying $A_j^2 = A_j$ and $A_j A_{j'} = 0$ for all $j \neq j'$. Let $a \neq 0$ be $n \times 1$ such that $A_j a = 0$, $j = 1, \dots, p$ and put $D' = (d_1, \dots, d_n)$. Then $z = X'a$ and $Q_j = X' A_j X$, $j = 1, \dots, p$, are independently distributed as $z \sim N_r(D'a, (a'a)V)$ and $Q_j \sim W_r(\rho_j, V, N_j)$, where $\rho_j = \text{rank } A_j$, and $N_j = D' A_j D V^{-1}$, $j = 1, \dots, p$.

3. Distributions of Statistics which are Invariant with Respect to Permutations

Let $x_1, \dots, x_n$ be a random sample from $\Pi_k(p_i; q_i(x))$. Let $g_t = g_t(x_i; j = 1, \dots, n)$, $t = 1, \dots, r$, be r statistics. If g_t , $t = 1, \dots, r$, are invariant with respect to permutations of $x_1, \dots, x_n$. Then, the following theorem provides the joint distribution of g_t , $t = 1, \dots, r$.

Theorem (3.1)

Let $f(g | a_i, i = 1, \dots, k) = f(g | a_i)$ be the joint conditional density of $g_t, t = 1, \dots, r$, given that the first a_1 sample points come from $q_1(x)$, the second a_2 sample points from $q_2(x)$, ..., and the last a_k sample points from $q_k(x)$. Then, the joint density of $g_t, t = 1, \dots, r$, is:

$$(3.1) \quad f(g) = \sum_{W_k(a_i; n)} \binom{n}{a_1, \dots, a_k} \left(\prod_{i=1}^k p_i^{a_i} \right) f(g | a_i),$$

Proof:

Given the condition that a_i of the sample points come from $q_i(x)$, $i = 1, \dots, k$ and noting the invariance of $g_t, t = 1, \dots, r$, the joint conditional distribution of $g_t, t = 1, \dots, r$ is then $f(g | a_i, i = 1, \dots, k)$. Since

$$\binom{n}{a_1, \dots, a_k} \prod_{i=1}^k p_i^{a_i}$$

is the probability of obtaining a sample with a_i sample points from $q_i(x)$, $i = 1, \dots, k$, so the result of the theorem.

QED

An Example:

As an application of the above theorem, we assume that $q_i(x)$ in $\Pi_k(p_i; q_i(x))$ is p -dimensional multivariate normal with mean μ_i and common covariance matrix Σ and proceed to derive the joint density of

$$G_1 = \frac{1}{n} \sum_{j=1}^n x_j = \bar{x} \quad \text{and} \quad G_2 = \sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})' = R.$$

Let a_i be the number of observations from $q_i(x)$, $i = 1, \dots, k$. Then $a_i \geq 0$ and

$\sum_{u=1}^k a_u = n$. Write x_j as x_{uv} if $a_u > 0$ and if x_j is the u -th observation from $q_u(x)$.

Then,

$$\bar{x} = \frac{1}{n} \sum_{j=1}^n x_j = \frac{1}{n} \sum_{j=1}^k a_u \bar{x}_u,$$

where $\bar{x}_u = \mathbf{0}$ if $a_u = 0$ and $\bar{x}_u = \frac{1}{a_u} \sum_{v=1}^{a_u} x_{uv}$ if $a_u \geq 1$.

Put $T_B = \sum_{u=1}^k a_u (\bar{x}_u - \bar{x})(\bar{x}_u - \bar{x})'$ and define $W_u = 0$ if $a_u = 0$ and

$$W_u = \sum_{v=1}^{a_u} (x_{uv} - \bar{x}_u) (x_{uv} - \bar{x}_u)' \text{ if } a_u \geq 1. \text{ Then}$$

$$R = \sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})' = \sum_{u=1}^k W_u + T_B.$$

Putting $\bar{\mu}_\omega = \frac{1}{n} \sum_{i=1}^k a_i \mu_i$, then it is readily seen that $\bar{x} \sim N_p(\bar{\mu}_\omega, \frac{1}{n} \Sigma)$ and $W = \sum_{u=1}^k W_u \sim W_{p(n-k), \Sigma}$, independently of $\bar{x}$. Further, applying Lemma 2, $T_B \sim W_p(k-1; \Sigma, \theta)$, independently of $\bar{x}$ and W , where

$$\theta = \left\{ \sum_{i=1}^k a_i (\mu_i - \bar{\mu}_\omega)(\mu_i - \bar{\mu}_\omega)' \right\} \Sigma^{-1}.$$

It follows that the conditional joint density $f(\bar{x}, R | a_i, i=1, \dots, k) = f(\bar{x}, R | a_i)$ of $\bar{x}$ and R given that a_i of the observations come from $q_i(x)$, $i = 1, \dots, k$, is:

$$f(\bar{x}, R | a_i) = N(\bar{x}; \bar{\mu}_\omega, \frac{1}{n} \Sigma) G_{n-1}(R; \Sigma, \theta),$$

a product of normal and non-central Wishart densities. Thus, the joint density $f(\bar{x}, R)$ of $\bar{x}$ and R is:

$$(3.2) \quad f(\bar{x}, R) = \frac{\Sigma}{W_k(a_i; n)} \binom{n}{a_1, \dots, a_k} \left(\prod_{i=1}^k P_i^{a_i} \right) N(\bar{x}; \bar{\mu}_\omega, \frac{1}{n} \Sigma) G_{n-1}(R; \Sigma, \theta),$$

This result is first given in Tan (1978).

Suppose that for each $i = 1, \dots, m$, y_{ij} , $j = 1, \dots, n$, is a random sample from $\Pi_k(p_u; q_u(y))$ and that the statistics $g_t = g_t(y_{ij}, i=1, \dots, m; j=1, \dots, n)$, $t = 1, \dots, r$, are invariant with respect to permutations on $(y_{i1}, \dots, y_{in})$ for all $i = 1, \dots, m$. Then we have the following theorem for the joint density of g_t , $t = 1, \dots, r$:

Theorem (3.2)

Let $f(g_t | s_{uv}) = f(g_t, t=1, \dots, r | s_{uv}, u=1, \dots, k, v=1, \dots, m)$ be the joint conditional density of g_t , $t=1, \dots, r$ given that, for each $v = 1, \dots, m$, the first s_{1v} observations of $(y_{v1}, \dots, y_{vn})$ come from $q_1(y)$ in $\Pi_k(p_u; q_u(y))$, the second s_{2v} observations from $q_2(y)$ in $\Pi_k(p_u; q_u(y))$, ..., the last s_{kv} observations from $q_k(y)$ in $\Pi_k(p_u; q_u(y))$. (Note that $\sum_{u=1}^k s_{uv} = n$, $v = 1, \dots, m$). Then the joint density $f(g_t, t=1, \dots, r)$ of g_t , $t = 1, \dots, r$, is:

$$(3.3) \quad f(g_t, t=1, \dots, r) = W_k(s_u; N) \binom{N}{s_1, \dots, s_k} \left(\prod_{u=1}^k P_u^{s_u} \right) W_m(s_{1v}; s_1) \dots \\ W_m(s_{kv}; s_k) \sum h(s_{uv}; s_u) f(g_t | s_{uv}),$$

where $N = mn$, and $s_u = \sum_{v=1}^m s_{uv}$, $u = 1, \dots, k$,

Proof

Consider the situation where s_{uv} of the observations $(y_{v1}, \dots, y_{vn})$ come from $q_u(y)$ in $\Pi_k(p_u; q_u(y))$, $u = 1, \dots, k$ for each $v = 1, \dots, m$.

Noting the invariance property, the joint conditional density of g_t , $t = 1, \dots, r$ is then $f(g_t | s_{uv})$. But now, $\sum_{u=1}^k s_{uv} = n$, $v = 1, \dots, m$; and $\sum_{v=1}^m s_{uv} = s_u$ is the total number of observations $(y_{ij}, i=1, \dots, m; j=1, \dots, n)$ from $q_u(y)$ in $\Pi_k(p_u; q_u(y))$, $u = 1, \dots, k$. Since $\prod_{v=1}^m \left\{ \binom{n}{s_{1v}, \dots, s_{kv}} \left(\prod_{u=1}^k P_u^{s_{uv}} \right) \right\} = \left(\prod_{u=1}^k P_u^{s_u} \right) \prod_{v=1}^m \binom{n}{s_{1v}, \dots, s_{kv}}$

is the probability of obtaining the sample as specified above, so the joint density of g_t , $t = 1, \dots, r$, is:

$$f(g_t, t=1, \dots, r) = W_k(s_u; N) \left\{ W_m(s_{1v}; s_1) \dots W_m(s_{kv}; s_k) \right\} \left(\prod_{u=1}^k P_u^{s_u} \right) \times \\ \left[\left\{ \prod_{v=1}^m \binom{n}{s_{1v}, \dots, s_{kv}} \right\} f(g_t | s_{uv}) \right] = W_k(s_u; N) \binom{N}{s_1, \dots, s_k} \left(\prod_{u=1}^k P_u^{s_u} \right) \times \\ \left\{ W_m(s_{1v}; s_1) \dots W_m(s_{kv}; s_k) \right\} h(s_{un}; s_u) f(g_t | s_{uv}).$$

An Example

Consider the model $y_{ij} = \mu + \alpha_i + \epsilon_{ij}$, $i = 1, \dots, m$, $j = 1, \dots, n$, where ϵ_{ij} , $i = 1, \dots, m$, $j = 1, \dots, n$ is a random sample from $\Pi_k(p_u; q_u(y))$ with $q_u(y)$ being multivariate normal with mean Δ_u satisfying $\sum_{u=1}^k p_u \Delta_u = 0$, common covariance matrix Σ and α_i are vectors of unknown parameters satisfying $\sum_{i=1}^m \alpha_i = 0$. Suppose that one is interested in testing the hypothesis $H_0: \alpha_i = 0$. The testing statistics usually involve the sum of squares matrices

$$R = n \sum_{i=1}^m (\bar{y}_{i.} - \bar{y}_{..})(\bar{y}_{i.} - \bar{y}_{..})' \quad \text{and}$$

$$T = \sum_{i=1}^m \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})(y_{ij} - \bar{y}_{i.})', \quad \text{where}$$

$$\bar{y}_{i.} = \frac{1}{n} \sum_{j=1}^n y_{ij} \quad \text{and} \quad \bar{y}_{..} = \frac{1}{m} \sum_{i=1}^m \bar{y}_{i.}.$$

Applying Theorem (3.2), we now proceed to derive the joint null distribution of R and T. Obviously, under $H_0: \alpha_i = 0$,

$$T = \sum_{i=1}^m \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})(y_{ij} - \bar{y}_{i.})' = \sum_{i=1}^m \sum_{j=1}^n (\mathbf{e}_{ij} - \bar{\mathbf{e}}_{i.})(\mathbf{e}_{ij} - \bar{\mathbf{e}}_{i.})'$$

and

$$R = R_{(e)} = n \sum_{i=1}^m (\bar{\mathbf{e}}_{i.} - \bar{\mathbf{e}}_{..})(\bar{\mathbf{e}}_{i.} - \bar{\mathbf{e}}_{..})', \quad \text{where } \bar{\mathbf{e}}_{i.} = \frac{1}{n} \sum_{j=1}^n \mathbf{e}_{ij}$$

and

$$\bar{\mathbf{e}}_{..} = \frac{1}{m} \sum_{i=1}^m \bar{\mathbf{e}}_{i.}, \text{ are invariant with respect to permutations of } (\mathbf{e}_{i1}, \dots, \mathbf{e}_{in}) \text{ for all } i = 1, \dots, m.$$

Hence, given that for each $v = 1, \dots, m$, s_{uv} of the observations $(y_{v1}, \dots, y_{vn})$ come from $q_u(y)$ in $\Pi_k(p_u; q_u(y))$, $u = 1, \dots, k$, by the result of the example in Theorem (3.1), the joint conditional density of $\mathbf{e}_i$.

and

$$T_{(i)} = \sum_{j=1}^n (\mathbf{e}_{ij} - \bar{\mathbf{e}}_{i.})(\mathbf{e}_{ij} - \bar{\mathbf{e}}_{i.})', \quad i = 1, \dots, m, \text{ is:}$$

$$\prod_{i=1}^m \{N(\mathbf{e}_{i.}; \Delta_{i(s)}, \frac{1}{n} \Sigma) G_{n-1}(T_{(i)}; \Sigma, \theta_i)\}, \text{ where}$$

$$\bar{\Delta}_{v(s)} = \frac{1}{n} \sum_{u=1}^k s_{uv} \Delta_u \text{ and}$$

$$\theta_v = \left\{ \sum_{u=1}^k s_{uv} (\Delta_u - \bar{\Delta}_{v(s)})(\Delta_u - \bar{\Delta}_{v(s)})' \right\} \Sigma^{-1}, \quad v = 1, \dots, m.$$

Making the transformation $T = \sum_{i=1}^m T_{(i)}$ and

$R_{(e)} = n \sum_{i=1}^m (\bar{e}_{i.} - \bar{e}_{..})(\bar{e}_{i.} - \bar{e}_{..})'$ and applying Lemma 2, we have then for the conditional joint density of $(R_{(e)}, T)$ given that s_{uv} of $(y_{v1}, \dots, y_{vn})$ comes from $q_u(y)$, $u = 1, \dots, k$ for all $v = 1, \dots, m$:

$$(3.4) \quad f(R_{(e)}, T | s_{uv}) = G_{k-1}(R_{(e)}; \Sigma, \Phi_a) G_{N-k}(T; \Sigma, \theta_w),$$

where

$$\begin{aligned} \Phi_a &= \{n \sum_{i=1}^m (\bar{\Delta}_{i(s)} - \bar{\Delta}_{..(s)})(\bar{\Delta}_{i(s)} - \bar{\Delta}_{..(s)})'\} \Sigma^{-1} \\ &= \frac{1}{n} \{\Delta L' (I_m - \frac{1}{m} J_m) L \Delta'\} \Sigma^{-1} \quad \text{with} \end{aligned}$$

$\bar{\Delta}_{..(s)} = \frac{1}{m} \sum_{i=1}^m \bar{\Delta}_{i(s)}$, $\Delta = (\Delta_1, \dots, \Delta_k)$, $L' = (s_{uv})$ and J_m a $m \times m$ matrix of 1's, and where

$$\begin{aligned} \theta_w &= \sum_{j=1}^m \theta_j = \left\{ \sum_{i=1}^k \sum_{j=1}^m s_{ij} (\Delta_i - \bar{\Delta}_{j(s)})(\Delta_i - \bar{\Delta}_{j(s)})' \right\} \Sigma^{-1} \\ &= \left\{ \Delta (D_s - \frac{1}{n} L' L) \Delta' \right\} \Sigma^{-1}, \quad \text{with} \end{aligned}$$

$$s_u = \sum_{v=1}^m s_{uv}, u = 1, \dots, k \quad \text{and}$$

$$D_s = \text{Diag}(s_1, \dots, s_k).$$

It follows that, under $H_0: \alpha_1 = 0$, the joint density of R and T is:

$$\begin{aligned} (3.5) \quad f(R, T) &= W_k(s_u; N) \left(s_1, \dots, s_k \right) \left(\prod_{u=1}^k p_u^{s_u} \right) W_m(s_{1v}; s_1) \dots W_m(s_{kv}; s_k) \times \\ &\quad h(s_{uv}; s_u) G_{m-1}(R; \Sigma, \Phi_a) G_{m(n-1)}(T; \Sigma, \theta_w), \end{aligned}$$

4. Some Distribution Results from Mixtures of Multivariate Normals

In Theorems (3.1) and (3.2), it is assumed that the statistics g_t , $t = 1, \dots, r$, are invariant with respect to permutations of the observations. In cases where this assumption does not hold, these theorems are in general not applicable. By assuming that $q_u(y)$ in

$\Pi_k(p_u; q_u(y))$ is a p-dimensional normal density with mean μ_u and common covariance matrix Σ , we provide, in this section, some distribution results for linear forms and quadratic forms. These results will then be used to derive the joint distribution of sample covariance matrices for a general linear hypothesis of multivariate regression models.

Now, applying Lemma 1 and simplifying, one has for the likelihood function $L(\mu_u, p_u, u=1, \dots, k, \Sigma | Y)$ of $(\mu_u, p_u, u=1, \dots, k, \Sigma)$ given $Y = (y_1, \dots, y_n)$:

$$(4.1) \quad L(\mu_u, p_u, u=1, \dots, k, \Sigma | Y) \\ = W_k(s_u; n) \left(\prod_{u=1}^k p_u^{s_u} \right) H_{k,n}(\Sigma_{uv}, s_u) \left(\frac{1}{\sqrt{2\pi}} \right)^{pn} |\Sigma|^{-n/2} \times \\ \exp \left\{ -\frac{1}{2} \text{tr} \Sigma^{-1} (Y - U\delta)(Y - U\delta)' \right\},$$

where

$$U = (\mu_1, \dots, \mu_k) \text{ and } \delta' = (\delta_1, \dots, \delta_k) \text{ with}$$

$$\delta_u' = (\delta_{u1}, \dots, \delta_{un}), u = 1, \dots, k.$$

Let $Z = Yb$, $Q = YAY'$ and $Q_i = YA_iY'$, $i = 1, \dots, r$, where A and A_i are symmetric matrices. Applying Lemma 2 to $L(\mu_u, p_u, u=1, \dots, k, \Sigma | Y)$, one has immediately the following two theorems:

Theorem (4.1)

If $Ab = 0$ and $A^2 = A$, then the joint density of Z and Q is:

$$(4.2) \quad g_1(z, Q) = W_k(s_u; n) \left(\prod_{u=1}^k p_u^{s_u} \right) H_{k,n}(\Sigma_{uv}, s_u) N(z; U\delta b, (b'b) \Sigma) \times \\ G_q(Q; \Sigma, \Phi),$$

where $q = \text{rank} A$ and $\Phi = U\delta A(U\delta)' \Sigma^{-1}$.

Theorem (4.2)

If $A_i b = 0$, $i = 1, \dots, r$ and $A_i A_j = 0$ for all $i \neq j$, then the joint density of $Z = Yb$ and $Q_i = YA_iY'$, $i = 1, \dots, r$, is:

$$(4.3) \quad g_2(\mathbf{z}, Q_i, i=1, \dots, r) = w_k(s_u; n) \left(\prod_{u=1}^k p_u^{s_u} \right) H_{k,n}(\delta_{uv}; s_u) N(\mathbf{z}; U\delta\mathbf{b}, (\mathbf{b}'\mathbf{b})\Sigma) \times \\ \left\{ \prod_{j=1}^r G_{q_j}(Q_j; \Sigma, \Phi_j) \right\},$$

where $q_j = \text{rank} A_j$ and $\Phi_j = U\delta A_j(U\delta)' \Sigma^{-1}$, $j = 1, \dots, r$.

Combining the results of the above two theorems, one has immediately the following theorem for the sampling distribution of linear and quadratic forms for mixtures of normals:

Theorem (4.3)

Suppose that $A = \sum_{j=1}^r A_j$, A_j symmetric and $A_j \mathbf{b} = \mathbf{0}$, $j = 1, \dots, r$. Consider the fol-

lowing four conditions:

$$(i) \quad A_j^2 = A_j, j = 1, \dots, r,$$

$$(ii) \quad A_i A_j = 0 \text{ for all } i \neq j,$$

$$(iii) \quad A^2 = A$$

and

$$(iv) \quad \text{rank} A = \sum_{j=1}^r \text{rank} A_j.$$

Then, if either (iii) and (iv) hold, or if any two conditions of (i), (ii) and (iii) hold, the joint density of $\mathbf{Z} = \mathbf{Y}\mathbf{b}$ and $Q_j = \mathbf{Y}A_j\mathbf{Y}'$, $j = 1, \dots, r$ is given by (4.3).

Proof:

Theorem (4.3) follows easily from Theorems (4.1) and (4.2) and a matrix result given in Rao (1975, pp. 28).

An Example:

As an application of Theorems (4.1) and (4.2) (or Theorem (4.3)) we consider the following multivariate regression model (For an application of this model, see Tiao and Zeller (1964)):

$$(4.4) \quad \begin{array}{ccccc} \mathbf{Y} & = & \theta & \mathbf{A} & + & \varepsilon \\ p \times N & & p \times q & q \times N & & p \times N \end{array}$$

where A is a given $q \times N$ matrix with $\text{rank} A = r (r \leq q)$, θ the $p \times q$ matrix of unknown parameters and ϵ the $p \times N$ matrix of random disturbances. As usual, it is assumed that the columns of ϵ is a random sample from a universe Π with mean 0. For model (4.4), suppose that one is interested in testing the general testable linear hypothesis $H_0: D\theta C = 0$, where C is $q \times s$ of $\text{rank} C = s$ and D $t \times p$ of $\text{rank} D = t$. Then, if Π is normal with covariance matrix Σ , the union-intersection test would reject H_0 if the largest root λ of $|H - \lambda E| = 0$ is greater than or equal to c_0 for some constant c_0 , where

$$H = (D\hat{\theta}C) \{C'(AA')^{-1}C\}^{-1} (D\hat{\theta}C)',$$

$E = D(Y - \hat{\theta}A)(Y - \hat{\theta}A)'D'$ and $\hat{\theta} = YA'(AA')^{-}$ with $(AA')^{-}$ being a g-inverse of AA' . (For proof, see Morrison (1977), pp. 177).

For determining c_0 , it is desirable to have the null joint distribution of H and E . If Π is normal with covariance matrix Σ , it is well-known that under H_0 , H and E are independently distributed as central Wishart, $H \sim W_t(\ell; D\Sigma D')$, where

$$\ell = \text{rank} \{A'(AA')^{-1}C \{C'(AA')^{-1}C\}^{-1} C'(AA')^{-1}A\} \text{ and}$$

$$E \sim W_t(n-r; D\Sigma D').$$

This is no longer the case if Π is not normal, however. For illustration, we assume that Π is a mixture of k multivariate normal distribution, i.e.

$$\Pi \sim \Pi_k(p_u; N(y; \mu_u, \Sigma)),$$

where μ_u satisfies $\sum_{u=1}^k p_u \mu_u = 0$, $u = 1, \dots, k$, and proceed to derive the joint null and nonnull distributions of H and E by applying Theorems (4.1) and (4.2) (or Theorem (4.3)).

Put $Z = DY$, $\phi = D\theta$ and $e = D\epsilon$. Then, $Z = \phi A + e$, where the columns of e is a random sample from $\Pi_k(p_u; N(e; \Delta_u, V))$, where

$$\Delta_u = D\mu_u, u = 1, \dots, k \text{ and } V = D\Sigma D'.$$

In terms of Z , we have then:

$$H = ZGZ' \text{ and } E = ZR_0Z',$$

where

$$G = A'(AA')^{-1}C \{C'(AA')^{-1}C\}^{-1} C'(AA')^{-1}A$$

and

$$R_0 = I_N - A' (AA')^{-1} A.$$

Obviously $GR_0 = 0$, $G^2 = G$ and $R_0^2 = R_0$. (Note that $AA' (AA')^{-1} A = A$ and $C' (AA')^{-1} AA' = C'$). By Theorem (4.3), the joint density of H and E is then given by:

$$f(H, E) = w_k(s_u; N) \left(\sum_{u=1}^k p_u^{s_u} \right) H_{k, N}(\delta_{uv}^{\sum} S_u) G_{\ell}(H; V, \Phi_1) G_{N-\ell}(E; V, \Phi_2),$$

where $\ell = \text{rank } G$,

$$\Phi_1 = \{(\phi A + \Delta\delta) G (\phi A + \Delta\delta)'\} V^{-1},$$

$$\Phi_2 = \Delta\delta R_0 (\Delta\delta)' V^{-1},$$

$$\Delta = (\Delta_1, \dots, \Delta_k) \text{ and } \delta = (\delta_{uv}), \text{ a } k \times N \text{ matrix.}$$

Note that $\phi AG = \phi C' \{C' (AA')^{-1} C\}^{-1} C' (AA')^{-1} A$, so, under H_0 ,
 $\Phi_1 = \Delta\delta G (\Delta\delta)' V^{-1}$

REFERENCES

- BOX, G.E.P. and TIAO, C.H. (1968), "A Bayesian approach to some outlier problems," *Biometrika* **55**, 119-129.
- DAVIS, A.W. (1976), "Statistical distributions in univariate and multivariate Edgeworth populations," *Biometrika* **63**:661-670.
- HYRENIUS, H. (1950), "Distribution of 'Student-Fisher's' t in samples from compound normal distribution," *Biometrika* **37**:429-442.
- HYRENIUS, H. (1952), "Sampling from bivariate nonnormal universes by means of compound normal distributions," *Biometrika* **39**:238-246.
- ITO, K. (1969), "On the effect of heteroscedasticity and nonnormality upon some multivariate test procedures," *Multivariate Analysis II*, edited by P.R. Krishnaiah, Academic Press, pp. 87-119.
- KRZANOWSKI, W.J. (1977), "The performance of Fisher's linear discriminant function under nonoptimal conditions," *Technometrics* **19**:191-200.
- LACHENBRUCH, P.A. et al (1977), "Robustness of the quadratic discriminant function to nonnormality," *Paper presented at the ASA and Biometric Society Annual Meeting, Chicago, ill.*, August 15-18, 1977.
- LAYARD, M.W.J. (1972), "Large sample tests for the equality of two covariance ma-

- trices," *Ann. Math. Statist.* **43**:123-141.
- LAYARD, M.W.J. (1973), "Robust large sample test for homogeneity of variances," *Jour. Amer. Statist. Assoc.* **68**:195-198.
- LAYARD, M.W.J. (1974), "A Monte Carlo comparison of tests for equality of covariance matrices," *Biometrika* **61**:461-465.
- LEE, A.F.S. and D'ANGOSTINO, R.B. (1976), "Levels of significance of some two-sample tests when observations are from compound normal distributions," *Comm. Statist.* **A5**(4):325-342.
- LEE, A.F.S. and GURLAND, J. (1977), "One-sample t-test when sampling from a mixture of normal distributions," *Ann. Statist.* **5**:803-807.
- MARDIA, K.V. (1971), "The effect of nonnormality on some multivariate tests and robustness to nonnormality in the linear model," *Biometrika* **58**:105-121.
- MARDIA, K.V. (1975), "Assessment of multinormality and the robustness of Hotelling's T^2 test," *Applied Statist.* **24**: 163-171.
- MORRISON, D.F. (1977), "*Multivariate Statistical Methods*," McGraw-Hill Book Company, N.Y.
- RAO, C.R. (1975), "*Linear Statistical Inference and its Applications*," John Wiley and Sons, Inc., N.Y.
- SUBRAHMANIAM, K. (1972), "On quadratic forms from mixtures of two normal populations," *South African Statist. J.* **6**:103-120.
- SUBRAHMANIAM, K. et al (1975), "On the robustness of some tests significance in sampling from a compound normal population," *J. Amer. Statist. Assoc.* **70**:435-438.
- TAN, W.Y. (1975), "On the biometrical analysis of quantitative genetics," *Botanic Bull.* **16**:67-82.
- TAN, W.Y. (1978), "On distributions of sample covariance matrices from a mixture of normal densities," *South African Statist. J.* **12**:47-55.
- TAN, W.Y. and CHANG, W.C. (1972), "Convolution approach to the analysis of quantitative characters in self-fertilized populations," *Biometrics* **28**:1073-1090.
- TIAO, C.H. and ALI, M.M. (1971), "Effect of nonnormality on inference about variance components," *Technometrics* **13**:635-650.
- TIAO, C.H. and ZELLNER, A. (1964), "On the Bayesian estimation of multivariate regression," *J. Roy. Statist. Society, Series B*, **26**:217-285.
- WATERNAUX, C.M. (1976), "Asymptotic distribution of the sample roots for a non-normal population," *Biometrika* **63**:639-646.
- WATERNAUX, C.M. (1977), "Principal components analysis in the nonnormal case," *Paper presented at the ASA and Biometric Society Meeting in Chicago, Ill., August 15-18, 1977*